Analytical benchmarks of the `tmm-core` package

Sixteen worked examples from first principles to ellipsometry

D. Redka[1]

[1] *Laser Center HM, Munich University of Applied Sciences HM*

(Dated: April 22, 2026)

The sixteen worked examples collected below validate the transfer-matrix method (TMM) implementation against closed-form analytical solutions wherever such solutions exist. The ordering runs from the single planar interface to absorbing multilayers and ellipsometric inversion, so that each successive geometry introduces one additional ingredient of the theory. Throughout, the code uses the sign convention in which $r_s = r_p$ at normal incidence; where a textbook formula is stated in the opposite sign convention, the discrepancy reduces to an overall sign of r_p and is flagged in the corresponding section. The time convention is $e^{-\mathrm{i}\omega t}$, and the branch $\mathrm{Im}(q_j) \geq 0$ is enforced for the complex out-of-plane wavenumber of every layer, so that evanescent and absorbing waves decay in the direction of propagation.

CONTENTS

I. Fresnel single interface 2

II. Fresnel with complex index 3

III. Total internal reflection and frustrated TIR 5

IV. Thin-film Airy formula 6

V. Half-wave and quarter-wave identities 7

VI. Single-layer antireflection coating 9

VII. Absorbing layer: energy conservation and Beer–Lambert limit 10

VIII. Beer–Lambert depth profile in a metal 11

IX. Metal reflectance of silver and gold 13

X. Distributed Bragg reflector 14

XI. Angle–wavelength map of a bandpass filter 15

XII. Fabry–Pérot finesse versus mirror reflectance 17

XIII. Symmetry and energy-conservation checks 19

XIV. Ellipsometric angles ψ and Δ 20

XV. Pseudo-dielectric function from ellipsometry 22

XVI. Pseudo-dielectric inversion on gold 24

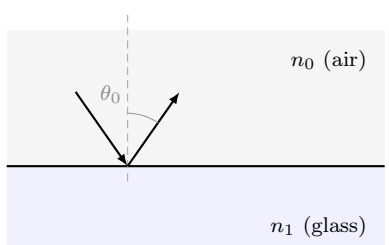
I. FRESNEL SINGLE INTERFACE

The first benchmark reduces the TMM to its simplest non-trivial case: a single planar interface between two semi-infinite half-spaces, without any finite layer at all. Light enters from air ($n_0 = 1.00$) onto a bulk crown-glass substrate ($n_1 = 1.50$) at the fixed wavelength $\lambda = 550\,\mathrm{nm}$, and the angle of incidence θ_0 is swept from 0 to 89.9° on a 901-point grid. Because there is no finite layer, the TMM is called with an empty thickness list, and the calculation collapses algebraically to the Fresnel amplitude coefficients

$$r_s = \frac{n_0 \cos\theta_0 - n_1 \cos\theta_1}{n_0 \cos\theta_0 + n_1 \cos\theta_1}, \tag{1}$$

$$r_p = \frac{n_1 \cos\theta_0 - n_0 \cos\theta_1}{n_1 \cos\theta_0 + n_0 \cos\theta_1}, \tag{2}$$

with Snell's law $n_0 \sin\theta_0 = n_1 \sin\theta_1$ fixing the refraction angle. The code uses the sign convention in which $r_s(0) = r_p(0)$ at normal incidence; the formulas (1)–(2) stated in the usual Wikipedia form give $r_s(0) = -r_p(0)$, so they differ from the code by an overall sign in r_p only, which leaves $|r_p|^2$ and hence the reflectance unchanged. The Brewster angle at which r_p vanishes follows from (2) as $\theta_B = \arctan(n_1/n_0) = 56.31°$.



The TMM reproduces the closed-form reflectance to $\max|\Delta R| \approx 4 \times 10^{-13}$ across the full sweep, and $|r_p(\theta_B)| \approx 3 \times 10^{-16}$ exactly at the Brewster angle. Both residuals are at the level of double-

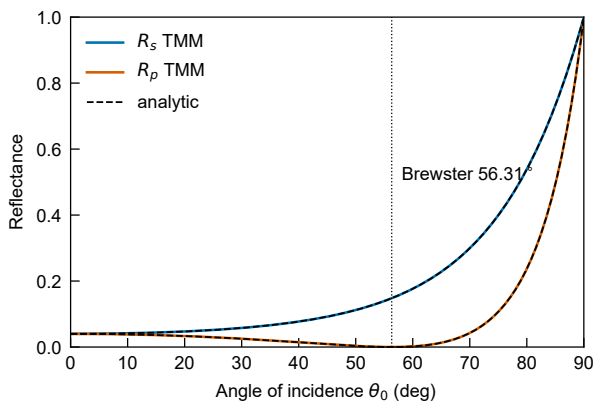


FIG. 1. Reflectance of an air/glass interface at 550 nm for s- and p-polarisation. TMM (colour) coincides with the closed-form Fresnel result (dashed) at machine precision. The Brewster angle is marked by the dotted vertical line.

precision floating-point noise, which confirms that the single-interface coefficients are evaluated correctly before any multilayer machinery is invoked.

II. FRESNEL WITH COMPLEX INDEX

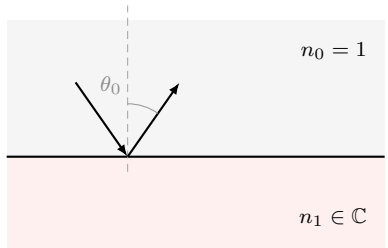
The second example extends the single-interface benchmark to a complex refractive index and adds the phase of the reflected amplitude as an independent check. Two cases are evaluated side by side at $\lambda = 550\,\mathrm{nm}$: air ($n_0 = 1.00$) against a non-absorbing glass substrate ($n_1 = 1.50$), and air against an absorbing Si-like substrate ($n_1 = 4.00 + 0.05\,\mathrm{i}$). The angle of incidence is sampled from 0 to 89.5° in 361 points. Because the substrate can be complex, Snell's law and the cosine that enters (1)–(2) must be evaluated on the physical branch,

$$\cos\theta_1 = \sqrt{1 - (n_0/n_1)^2 \sin^2\theta_0}, \qquad \mathrm{Im}(\cos\theta_1) \geq 0, \tag{3}$$

so that the transmitted wave decays into the absorbing half-space. At normal incidence the two polarisations become physically indistinguishable and the code returns

$$r_s(0) = r_p(0) = \frac{n_0 - n_1}{n_0 + n_1}, \tag{4}$$

whereas the textbook formulas written in the opposite sign convention give $r_s(0) = -r_p(0)$; the sign flip is applied once when overlaying the analytic reference on the TMM curve.



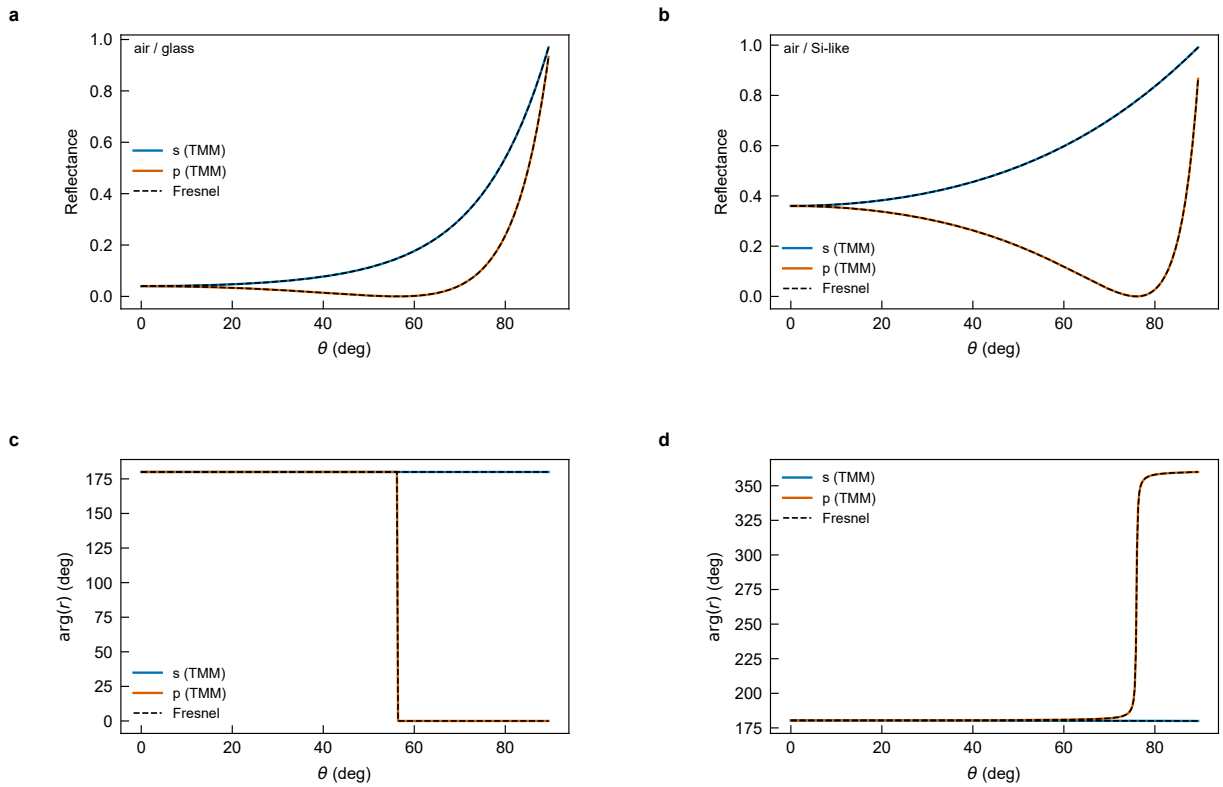


FIG. 2. TMM versus closed-form Fresnel for a real (glass) and a complex (Si-like) substrate. Panels (a,b) show $R_s(\theta)$ and $R_p(\theta)$; panels (c,d) show $\arg r_s$ and $\arg r_p$ over the same angular range.

With the sign convention fixed, amplitudes and phases coincide in all four panels at $\max|\Delta R| \approx 6\times10^{-14}$. The dielectric case shows the expected sharp Brewster zero and π phase jump in r_p, while the absorbing substrate replaces both by the smooth phase walk of Fig. 2(d). The agreement across real and complex substrates validates the branch-cut choice of (3), which is the single numerical subtlety that separates the complex case from the real one.

III. TOTAL INTERNAL REFLECTION AND FRUSTRATED TIR

The third example probes the same single-interface machinery beyond the critical angle and then promotes the geometry to a three-layer stack in order to resolve the evanescent tunnelling known as frustrated total internal reflection (FTIR). The incident medium is glass ($n_1 = 1.50$) and the transmitted medium is air ($n_2 = 1.00$) at $\lambda = 550\,\mathrm{nm}$, so the critical angle is $\theta_c = \arcsin(n_2/n_1) = 41.81°$. Above θ_c no real transmission angle exists; the normal component of the wavevector in air becomes imaginary, and unitarity demands

$$R(\theta > \theta_c) = 1. \tag{5}$$

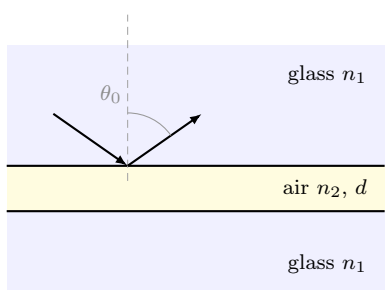
Panel (a) of Fig. 3 plots $|1 - R|$ for s- and p-polarisation on a logarithmic scale across the forbidden window $\theta_c < \theta < 89°$. Panel (b) compares the unwrapped phase of r_s to the textbook closed form

$$\tan\left(\frac{\delta_s}{2}\right) = \frac{\sqrt{\sin^2\theta - (n_2/n_1)^2}}{\cos\theta}. \tag{6}$$

Panel (c) inserts a finite air gap of variable thickness d between two glass half-spaces, so the stack becomes glass / air / glass with a single finite layer. Above θ_c the wave in the gap is evanescent, with imaginary out-of-plane wavenumber $q_2 = \mathrm{i}\kappa$ and $\kappa = k_0\sqrt{n_1^2 \sin^2\theta - n_2^2}$. The three-layer Airy transmittance then reads

$$t = \frac{t_{12}\, t_{21}\, e^{\mathrm{i}q_2 d}}{1 - r_{12}^2\, e^{2\mathrm{i}q_2 d}}, \qquad T = |t|^2, \tag{7}$$

which is evaluated at $\theta = 60°$ for gap thicknesses from 10 nm to 600 nm and overlaid on the TMM result.



Above θ_c the TMM returns unit reflectance to $|1 - R| \approx 9 \times 10^{-16}$ for both polarisations, and the reflection phase tracks (6) to the same tolerance. The FTIR curve agrees with the closed-form Airy transmittance at $\max|\Delta T| \approx 7 \times 10^{-16}$ down to sub-nanometre gaps. The transmittance follows the pure exponential $e^{-2\kappa d}$ for $d \gg \kappa^{-1}$, but deviates visibly from it below $d \lesssim 100\,\mathrm{nm}$; in that near-gap regime the Fresnel prefactors of (7) matter, and the TMM tracks those corrections exactly.

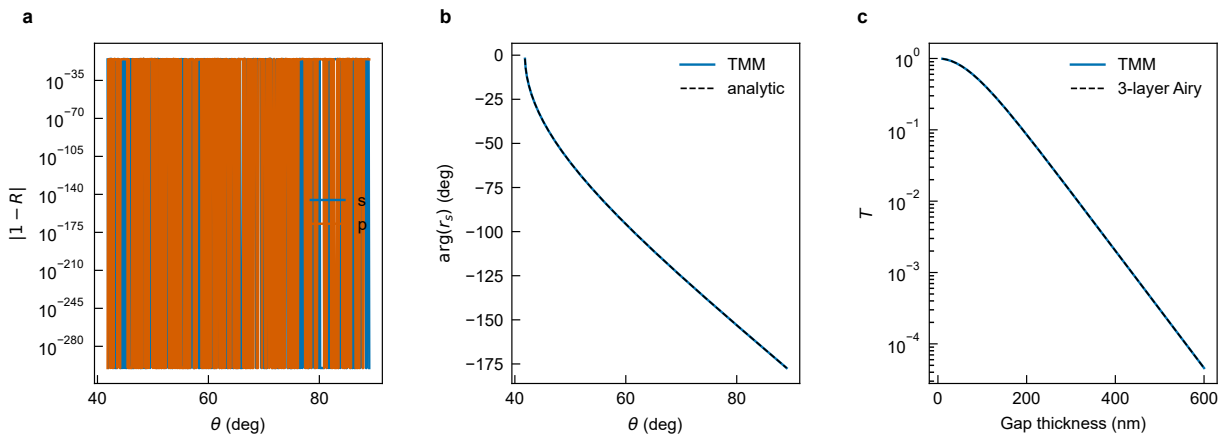


FIG. 3. (a) $|1 - R|$ above θ_c for s and p polarisation. (b) Phase of r_s versus the closed form (6). (c) Frustrated-TIR transmittance $T(d)$ at 60° against the exact three-layer Airy expression (7).

IV. THIN-FILM AIRY FORMULA

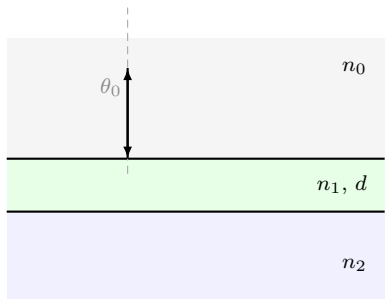
The fourth example pushes the benchmark from two to three media by placing a single finite film between two half-spaces and comparing the TMM against the closed-form Airy formula at normal incidence. Three configurations are run in parallel, sweeping the wavelength from 300 nm to 1000 nm:

- Case (a): air ($n_0 = 1$) / lossless film ($n_1 = 2.0$) / glass ($n_2 = 1.5$), film thickness $d = 100$ nm;
- Case (b): air / absorbing film ($n_1 = 3.0 + 0.05\,\mathrm{i}$) / glass, $d = 500$ nm;
- Case (c): air / absorbing film ($n_1 = 3.0 + 0.05\,\mathrm{i}$) / absorbing substrate ($n_2 = 3.11 + 4.89\,\mathrm{i}$), $d = 500$ nm.

Case (a) is purely interferometric, case (b) adds weak absorption in the film, and case (c) makes both the film and the substrate complex, which tests the branch-cut convention of both the film wavenumber and the substrate Fresnel coefficient simultaneously. The analytic reference is the Airy formula for a single film,

$$r = \frac{r_{01} + r_{12}\, e^{2\mathrm{i}\beta}}{1 + r_{01}\, r_{12}\, e^{2\mathrm{i}\beta}}, \qquad \beta = \frac{2\pi n_1 d}{\lambda} \cos\theta_1, \tag{8}$$

with r_{jk} the single-interface amplitudes of (1)–(2). The positive sign in $e^{+2\mathrm{i}\beta}$ is dictated by the $e^{-\mathrm{i}\omega t}$ time convention and is the sign that makes (8) consistent with the Fresnel amplitudes used by the TMM.



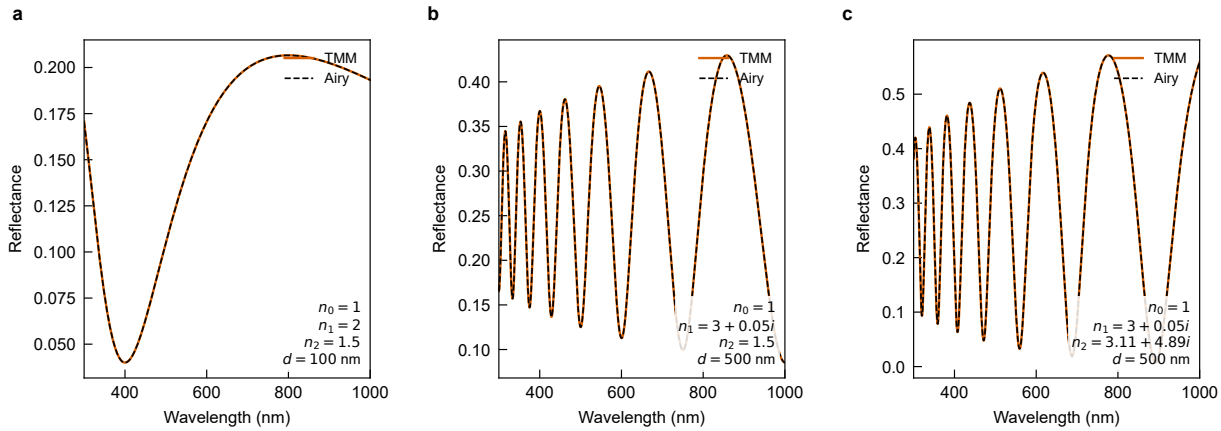


FIG. 4. Reflectance of a single film versus wavelength for the three cases listed in the text. TMM (solid colour) overlays the Airy formula (dashed black) in each panel.

Across all three cases the TMM and Airy curves coincide at $\max|R_{\mathrm{TMM}} - R_{\mathrm{Airy}}| \approx 4 \times 10^{-15}$, independent of whether the film or the substrate is absorbing. The residual is dominated by the catastrophic cancellation that enters (8) when the numerator is nearly zero, and it stays at the level of floating-point noise throughout the scan. Case (a) shows the textbook equally-spaced Airy fringes, case (b) damps the fringe contrast towards shorter wavelengths as the film absorption becomes significant, and case (c) adds a smooth background from the complex substrate Fresnel coefficient on which the attenuated fringes ride.

V. HALF-WAVE AND QUARTER-WAVE IDENTITIES

Two classical design identities of single-layer interferometry are checked at the design wavelength $\lambda_0 = 550\,\mathrm{nm}$. Both use a three-layer stack (incident medium, film, substrate) at normal incidence and are scanned across 400–800 nm for context.

The first panel realises the half-wave *invisible* layer: a lossless film of index $n_f = 2.0$ is placed

on glass ($n_s = 1.5$) in air ($n_0 = 1.0$), with thickness

$$d_{\mathrm{HW}} = \frac{\lambda_0}{2n_f} = 137.5\,\mathrm{nm}. \tag{9}$$

At λ_0 the film introduces a round-trip phase of 2π, so the internal Airy factor in (8) reduces to unity and the reflectance of the stack at that wavelength collapses to the bare-substrate value,

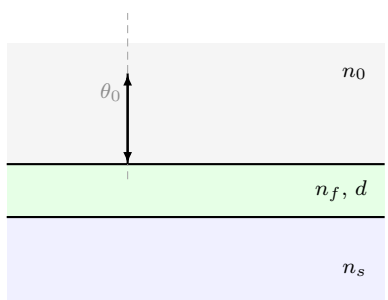
$$R(\lambda_0) = \left(\frac{n_0 - n_s}{n_0 + n_s}\right)^2. \tag{10}$$

The second panel realises the index-matched quarter-wave antireflection (AR) film: the film index is chosen as the geometric mean $n_f = \sqrt{n_0 n_s} \approx 1.225$, the thickness is

$$d_{\mathrm{QW}} = \frac{\lambda_0}{4n_f}, \tag{11}$$

and (8) then gives exact destructive interference at λ_0,

$$R(\lambda_0) = 0. \tag{12}$$



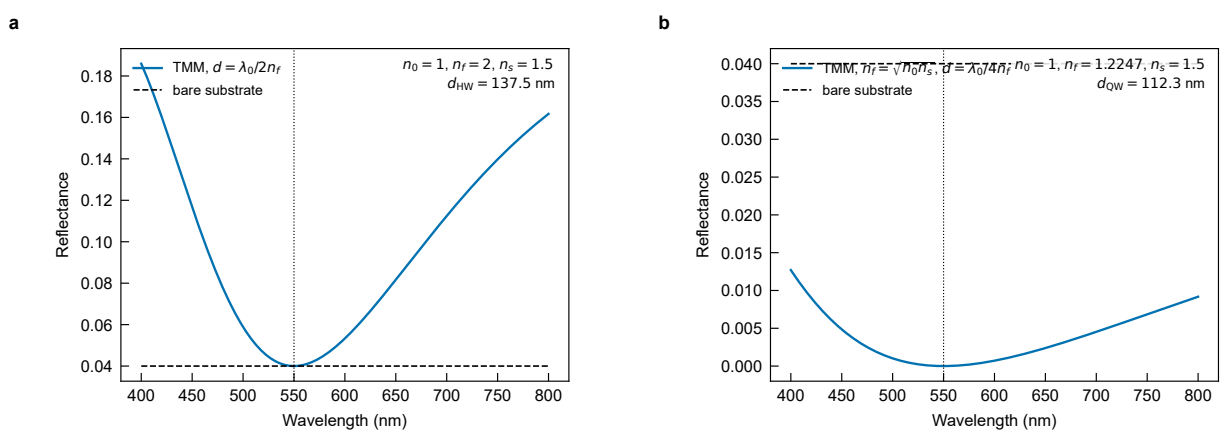


FIG. 5. (a) Half-wave invisible film returns the bare-substrate reflectance at λ_0. (b) Index-matched quarter-wave antireflection film gives zero reflectance at λ_0 and a smooth V-shape across the scan.

At λ_0 the half-wave film coincides with the bare substrate to $|R_{\mathrm{film}}(\lambda_0) - R_{\mathrm{bare}}| \approx 3 \times 10^{-17}$, and the index-matched quarter-wave film reaches $R(\lambda_0) \approx 6 \times 10^{-33}$; the latter is a squared amplitude

and therefore registers the reflection amplitude itself vanishing at the level of $\sqrt{6 \times 10^{-33}} \sim 8 \times 10^{-17}$, i.e. pure floating-point zero. The two identities constrain the Airy phase and the Fresnel zero condition independently, so their simultaneous satisfaction is a stronger check than either one alone.

VI. SINGLE-LAYER ANTIREFLECTION COATING

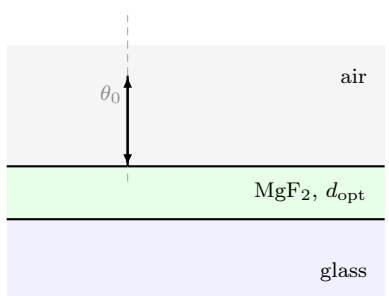
Beyond the idealised index-matched case of the previous section, the quarter-wave AR formula predicts a finite residual reflectance whenever the film index is not exactly $\sqrt{n_0 n_s}$. The benchmark here is a magnesium-fluoride film ($n_f = 1.38$) on crown glass ($n_s = 1.52$) in air ($n_0 = 1.00$) at the design wavelength $\lambda_0 = 550\,\mathrm{nm}$. The analytical optimum thickness and residual reflectance are

$$d_{\mathrm{opt}} = \frac{\lambda_0}{4 n_f} \approx 99.6\,\mathrm{nm}, \tag{13}$$

$$R_{\min} = \left(\frac{n_0 n_s - n_f^2}{n_0 n_s + n_f^2} \right)^2 . \tag{14}$$

$R_{\min}$ is the exact reflectance of the stack at λ_0 when $d = d_{\mathrm{opt}}$; it vanishes only in the index-matched limit $n_f^2 = n_0 n_s$.

Two independent scans are performed: panel (a) fixes the wavelength at λ_0 and sweeps the film thickness from 0 to 300 nm on a 601-point grid, locating the analytic minimum; panel (b) fixes the thickness at d_{opt} and sweeps the wavelength across 400–700 nm, overlaying the bare-substrate reflectance for reference.



The minimum of the TMM thickness scan falls exactly on (14) with $|R_{\mathrm{TMM}}(d_{\mathrm{opt}}, \lambda_0) - R_{\min}| \approx 7 \times 10^{-18}$, and the wavelength scan shows the familiar broadband V-shape touching the analytic $R_{\min}$ only at the design wavelength. The residual reflectance (here $R_{\min} \approx 1.3 \times 10^{-2}$) is a pure consequence of the index mismatch between MgF_2 and the glass–air geometry; broadening the AR spectrally requires multilayer designs, which are addressed by the Bragg and bandpass examples below.

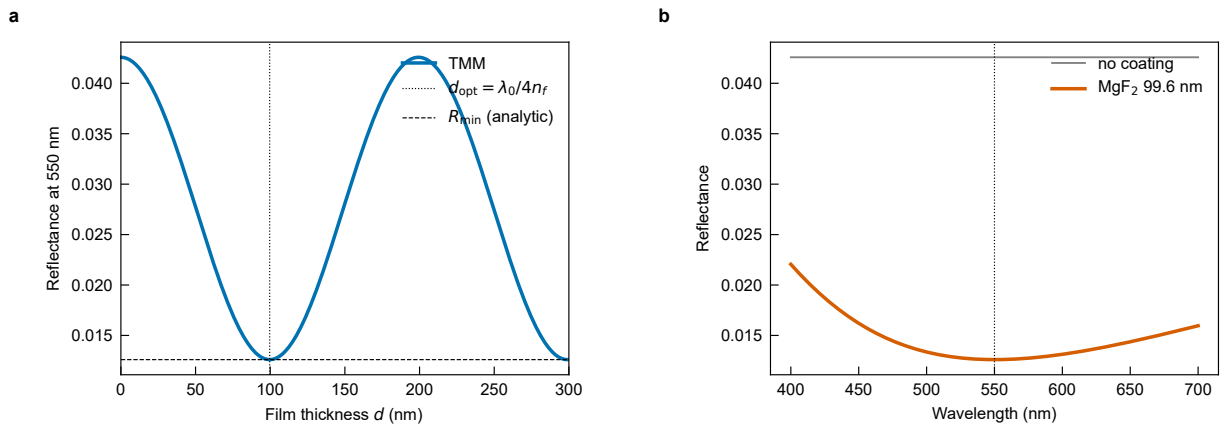


FIG. 6. (a) Reflectance as a function of film thickness at $\lambda_0 = 550\,\mathrm{nm}$, touching the analytic $R_{\min}$ at $d = d_{\mathrm{opt}}$. (b) Reflectance against wavelength at the optimum thickness, with the bare-substrate curve overlaid for comparison.

VII. ABSORBING LAYER: ENERGY CONSERVATION AND BEER–LAMBERT LIMIT

With absorption present in a finite layer, the reflectance and the transmittance no longer sum to unity; the missing fraction is deposited in the film itself. This example verifies, first, that the TMM partitions the incident power correctly ($R + T + A = 1$) and, second, that the transmittance approaches the expected Beer–Lambert exponential for optically thick films. The stack is an amorphous-silicon-like film ($n_f = 4.0 + 0.5\,\mathrm{i}$) on glass ($n_s = 1.52$) in air ($n_0 = 1.00$) at $\lambda = 550\,\mathrm{nm}$.

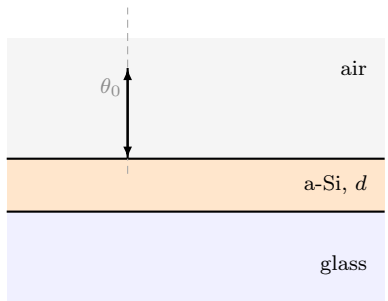
Panel (a) sweeps the film thickness from 0 to 400 nm in 801 points. The TMM returns the reflectance R, the transmittance T and the per-layer absorption A (here a single scalar because there is only one finite layer). The three components oscillate through Airy fringes as d grows, but energy conservation

$$R + T + A = 1 \tag{15}$$

must hold at every point. Panel (b) extends the scan out to 2000 nm, where the Airy ripple in $T(d)$ decays into the Beer–Lambert asymptote

$$T(d) \longrightarrow T_{\mathrm{if}}\, e^{-\alpha d}, \qquad \alpha = \frac{4\pi\ \mathrm{Im}(n_f)}{\lambda}, \tag{16}$$

with the interface transmittance $T_{\mathrm{if}} = (1 - R_{\mathrm{front}})(1 - R_{\mathrm{back}})$ built from the two single-interface reflectances. At $\lambda = 550\,\mathrm{nm}$ the numerical values are $\alpha \approx 1.14 \times 10^{-2}\,\mathrm{nm}^{-1}$, i.e. a penetration depth $1/\alpha \approx 87.5\,\mathrm{nm}$.



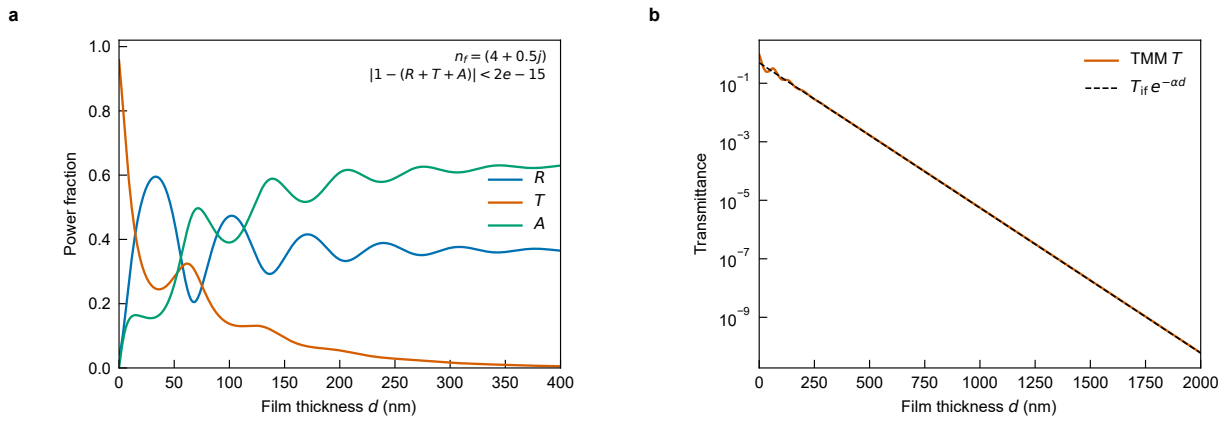


FIG. 7. (a) Reflectance R, transmittance T and per-layer absorption A as a function of film thickness. (b) $T(d)$ on a logarithmic scale out to 2000 nm, with the Beer–Lambert envelope $T_{\rm if}\,e^{-\alpha d}$ as a reference.

Equation (15) is satisfied to $\max|1-(R+T+A)| \approx 2\times10^{-15}$ over the whole thickness scan, which confirms the per-layer absorption formula derived from the Poynting flux balance. For $d \gtrsim 1/\alpha$ the oscillatory Fabry–Pérot modulation of $T(d)$ dies out, and beyond a few absorption lengths the TMM curve locks onto the exponential envelope of (16) without adjustable parameters. The transition from coherent interference to the Beer–Lambert regime is therefore reproduced smoothly by the same formalism.

VIII. BEER–LAMBERT DEPTH PROFILE IN A METAL

The per-layer absorption returned by the TMM can be used to resolve the spatial deposition of optical power with arbitrary resolution. This benchmark slices a gold half-space into $N = 100$ virtual layers of $dz = 2\,\text{nm}$ each, followed by a bulk gold half-space as substrate, so the total thickness (200 nm) is deep enough that the reflectance has converged to the bulk Fresnel value. The gold refractive index at $\lambda = 550\,\text{nm}$ is taken from the Johnson–Christy tabulation as $n_{\rm Au} =$

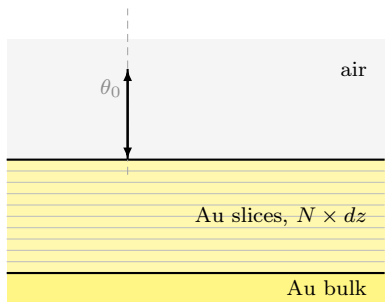
$0.425 + 2.367\,\mathrm{i}$. Because all N slices carry the same index as the substrate, the slicing is a pure reparametrisation of a continuous absorber.

With the field entering the metal at intensity $I_{\mathrm{in}} = 1 - R$ and the absorption coefficient $\alpha = 4\pi\ \mathrm{Im}(n_{\mathrm{Au}})/\lambda$, the differential and cumulative Beer–Lambert expressions are

$$\frac{\mathrm{d}A}{\mathrm{d}z} = \alpha\, I_{\mathrm{in}}\, e^{-\alpha z}, \tag{17}$$

$$A(< z) = I_{\mathrm{in}}\big(1 - e^{-\alpha z}\big). \tag{18}$$

Panel (a) compares the TMM per-layer absorption A_j, evaluated at the midpoint depth $z_j = (j - 1/2)\,dz$, against $\alpha\, dz\,(1 - R)\, e^{-\alpha z_j}$. Panel (b) overlays the cumulative sum $\sum_{k \le j} A_k$ on the closed form (18).



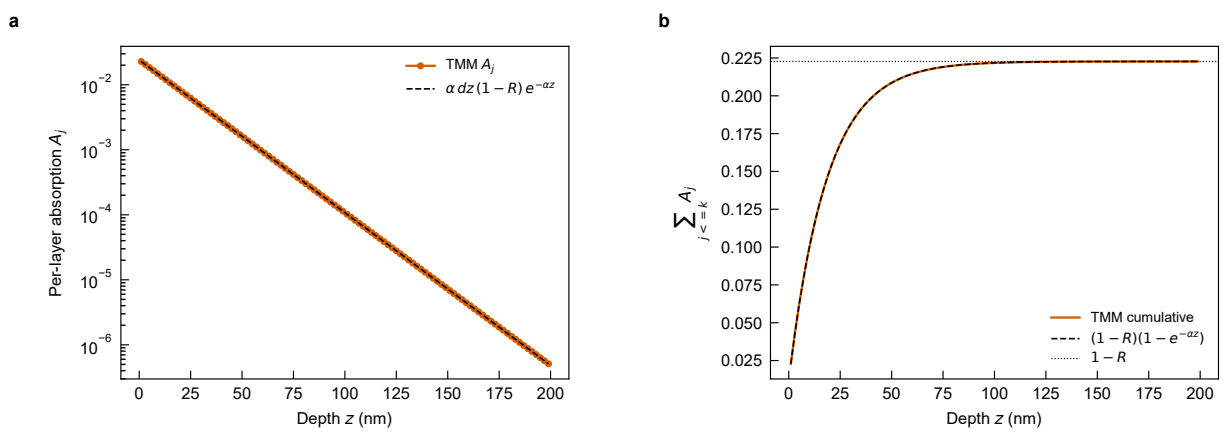


FIG. 8. (a) Per-layer absorption A_j at slice midpoints, TMM versus the Beer–Lambert prediction. (b) Cumulative absorption up to depth z, TMM against the closed form (18).

The cumulative curve matches (18) to $\max|A_{\mathrm{cum,TMM}} - A_{\mathrm{cum,BL}}| \approx 1 \times 10^{-15}$, which confirms the total energy partition of the TMM. The differential form (17) shows a residual of order 5×10^{-4} near the surface; this is not a methodological bias but the finite-difference error $\mathcal{O}(\alpha^2\, dz^2)$ inherent to approximating $\int_{z_j - dz/2}^{z_j + dz/2} e^{-\alpha z}\,\mathrm{d}z$ by the midpoint value times dz. Halving the slice thickness to $1\,\mathrm{nm}$ reduces the residual to $\sim 1 \times 10^{-4}$, consistent with the quadratic scaling.

IX. METAL REFLECTANCE OF SILVER AND GOLD

This example uses experimental optical constants instead of fabricated values. The refractive index of silver and gold is taken from the Johnson–Christy tabulation sampled at the ten wavelengths 400, 450, 500, 550, 600, 650, 700, 800, 900, 1000 nm, and linearly interpolated on a 601-point grid between 400 and 1000 nm. The stack is reduced to two half-spaces, air against metal, and the reflectance is evaluated at normal incidence for both materials.

For a bare interface the s- and p-polarisations coincide, and the normal-incidence reflectance follows directly from the Fresnel coefficients as

$$R = \frac{(n-1)^2 + k^2}{(n+1)^2 + k^2}, \tag{19}$$

with $N = n + \mathrm{i}k$ the complex refractive index at the given wavelength. Equation (19) is the strong reference against which the TMM result is compared: any disagreement would indicate an error in the branch-cut choice for complex N or in the evaluation of the single-interface Fresnel coefficient at $\theta_0 = 0$.

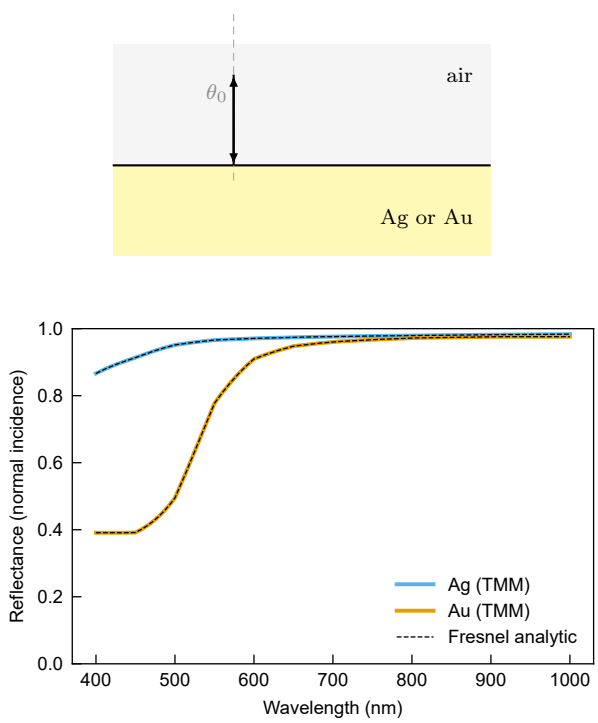


FIG. 9. Reflectance of bulk silver and bulk gold from 400–1000 nm at normal incidence. TMM (solid) is overlaid on the closed-form Fresnel expression (19) evaluated from the same tabulated $N(\lambda)$.

The TMM reproduces (19) at $\max|\Delta R| \approx 1 \times 10^{-15}$ for silver and 9×10^{-16} for gold. Silver remains above $R \approx 0.95$ across the whole visible and near-infrared range; gold shows the characteristic

absorption edge that pulls the reflectance down below 500 nm, with the reflectance rising again and saturating near unity in the red and near-infrared. The identity of the two curves to floating-point precision confirms that tabulated complex indices are ingested and propagated consistently by the code.

X. DISTRIBUTED BRAGG REFLECTOR

The first genuine multilayer benchmark is a quarter-wave distributed Bragg reflector (DBR). The stack alternates a high-index and a low-index quarter-wave layer and is repeated N times on a glass substrate. The materials are TiO_2 ($n_H = 2.35$) and SiO_2 ($n_L = 1.38$) on glass ($n_s = 1.52$) in air ($n_0 = 1.00$), with the design wavelength fixed at $\lambda_0 = 550\,\text{nm}$. Each layer thickness equals one optical quarter-wavelength, so $d_H = \lambda_0/(4n_H) \approx 58.5\,\text{nm}$ and $d_L = \lambda_0/(4n_L) \approx 99.6\,\text{nm}$. The refractive index list fed to the TMM is

$$(n_0,\ n_H,\ n_L,\ n_H,\ n_L,\ \ldots,\ n_H,\ n_L,\ n_s),$$

with the (n_H, n_L) block repeated N times, and the thickness list is the corresponding $(d_H, d_L)^N$ sequence.

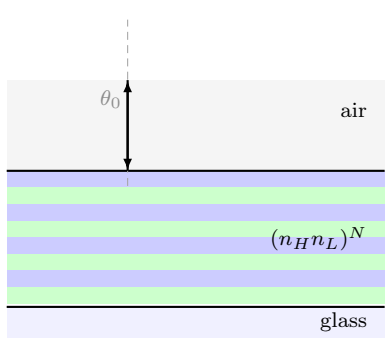
Two analytical results frame the behaviour. The width of the high-reflectance stopband centred at λ_0 is

$$\frac{\Delta\lambda}{\lambda_0} = \frac{4}{\pi}\arcsin\left(\frac{n_H - n_L}{n_H + n_L}\right), \tag{20}$$

and the peak reflectance at λ_0, with the effective substrate admittance $Y_{\text{eff}} = (n_H/n_L)^{2N}\, n_s$ for the stack terminated in n_L / substrate, is

$$R_{\text{peak}} = \left(\frac{n_0 - Y_{\text{eff}}}{n_0 + Y_{\text{eff}}}\right)^2. \tag{21}$$

Panel (a) plots $R(\lambda)$ across 400–800 nm for $N = 2, 5, 10$ pairs with the stopband from (20) shaded. Panel (b) sweeps N from 1 to 15 and compares the TMM peak reflectance at λ_0 with (21).



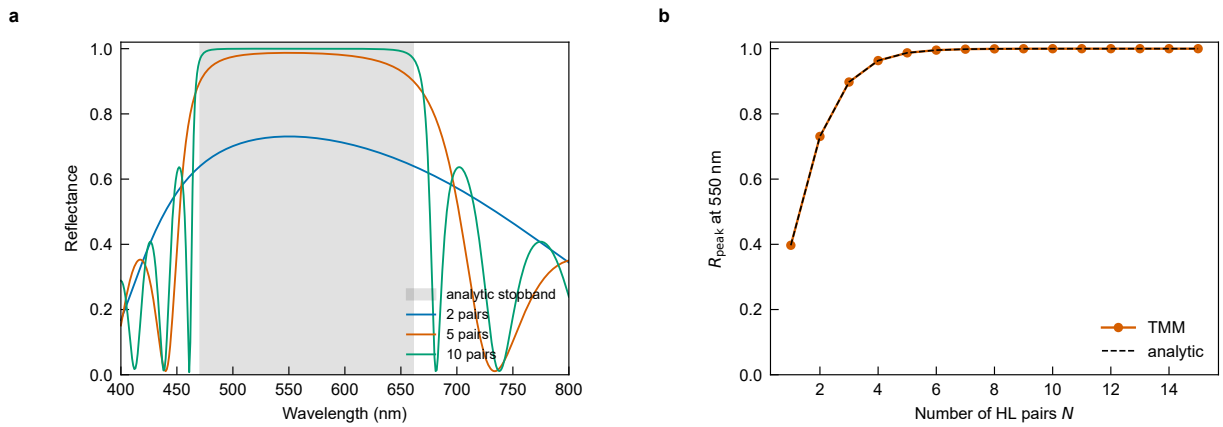


FIG. 10. (a) Reflectance of the DBR for $N = 2, 5, 10$ pairs on glass; the shaded band marks the analytic stopband (20). (b) Peak reflectance at λ_0 against the admittance formula (21).

The TMM peak reflectance tracks the admittance formula (21) over $N = 1 \ldots 15$ with $|\Delta R_{\text{peak}}| \approx 1 \times 10^{-15}$, and the high-reflectance window in panel (a) falls inside the analytic stopband at all pair counts. For $N \gtrsim 10$ the reflectance saturates above 99.99% at the design wavelength, and the stopband edges become progressively sharper with increasing N; outside the stopband the curves show the usual ripple from the finite number of periods, which the TMM reproduces in amplitude and phase.

XI. ANGLE–WAVELENGTH MAP OF A BANDPASS FILTER

This example generalises the DBR benchmark to a two-dimensional scan in wavelength and angle of incidence, using the simplest optical layout that exhibits a narrow transmission passband: a single half-wave cavity placed between two identical quarter-wave mirrors. Explicitly, the refractive index list fed to the TMM at design wavelength $\lambda_0 = 550\,\text{nm}$ is

$$\underbrace{\text{air}}_{n_0=1.00}\ \underbrace{(n_H\, n_L)^4}_{\text{mirror 1}}\ \underbrace{n_H}_{\text{spacer}}\ \underbrace{(n_L\, n_H)^4}_{\text{mirror 2}}\ \underbrace{\text{glass}}_{n_s=1.52}\,,$$

with $n_H = 2.35$ and $n_L = 1.38$. The seventeen finite layers carry the thicknesses $(d_H, d_L)^4$ for the first mirror, d_{cav} for the spacer, and $(d_L, d_H)^4$ for the second mirror, where

$$d_H = \frac{\lambda_0}{4n_H}, \quad d_L = \frac{\lambda_0}{4n_L}, \quad d_{\text{cav}} = \frac{\lambda_0}{2n_H}. \tag{22}$$

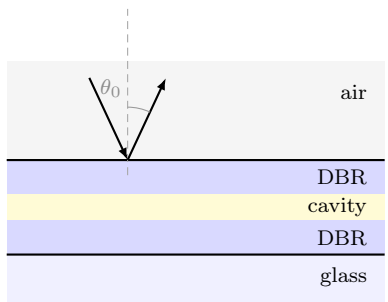
The spacer is therefore a *half*-wave high-index layer, which introduces a sharp transmission resonance at λ_0 sitting inside the mirror stopband. The high-reflectance blocks on either side of the

cavity give the filter its narrow-band character; the cavity phase alone sets the position of the passband.

The map scans λ from 500 to 600 nm on 201 points and the angle of incidence θ_0 from 0 to 40° on 41 points, for both s- and p-polarisation. At oblique incidence the resonance shifts to shorter wavelengths because the round-trip phase in the cavity picks up a factor of $\cos\theta_{\text{cav}}$; a first-order estimate treats the cavity as a single effective medium with index $n_{\text{eff}} \simeq \sqrt{n_H n_L}$, giving the textbook passband shift

$$\lambda(\theta) = \lambda_0 \sqrt{1 - \frac{\sin^2\theta_0}{n_{\text{eff}}^2}}. \tag{23}$$

Equation (23) is the standard approximation for thin-film bandpass filters and is accurate to a few per mille for modest angular excursions; it is plotted as a dashed red curve on top of both maps.



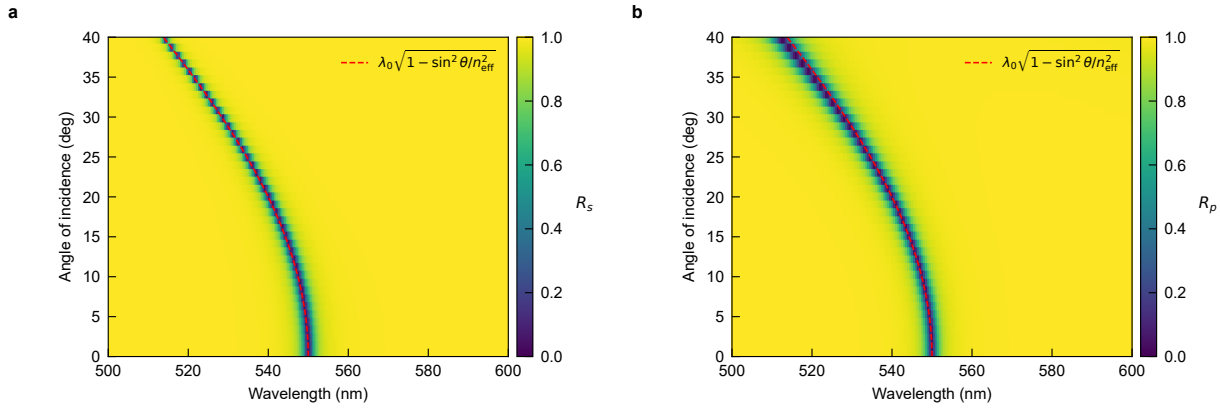


FIG. 11. Two-dimensional reflectance map $R(\theta_0, \lambda)$ of the bandpass filter for s (a) and p (b) polarisation. The dark vertical streak is the narrow cavity resonance, seen here as a dip in reflectance; the dashed red curve is the analytic passband shift of equation (23) with $n_{\text{eff}} = \sqrt{n_H n_L}$.

The analytic curve tracks the minimum of $R(\theta_0, \lambda)$ in both maps without adjustable parameters, which validates the angular dispersion of the stack up to the approximation used in n_{eff}. A second

feature emerges at oblique incidence: the s and p maps are no longer identical. The p map shows a shallower and slightly broader dip at large θ_0 because each interface carries a different Fresnel coefficient for p polarisation, which reduces the effective mirror reflectance and broadens the cavity line. This polarisation splitting of the passband is the exact quantity that tilted-incidence bandpass filters are designed around, and it is reproduced here from first principles without any effective-medium fit parameter.

XII. FABRY–PÉROT FINESSE VERSUS MIRROR REFLECTANCE

The previous example introduced the single-cavity bandpass geometry as a sharp feature on top of a high-reflectance background. Here the same building block is evaluated quantitatively: the transmittance spectrum of a symmetric Fabry–Pérot etalon is simulated for a range of mirror reflectances, and the finesse extracted from the TMM is compared to the analytic Airy prediction. The design wavelength is $\lambda_0 = 1000\,\mathrm{nm}$.

The etalon is built from two identical quarter-wave mirrors surrounding a long air cavity. With $n_H = 2.35$, $n_L = 1.46$ (SiO_2-like), $n_0 = n_{\mathrm{cav}} = 1.00$, and layer thicknesses $d_H = \lambda_0/(4n_H) \approx 106.4\,\mathrm{nm}$, $d_L = \lambda_0/(4n_L) \approx 171.2\,\mathrm{nm}$, the index list fed to the TMM is

$$n_0\ (n_H\, n_L)^N\ n_{\mathrm{cav}}\ (n_L\, n_H)^N\ n_0,$$

with matching thicknesses $(d_H, d_L)^N,\ d_{\mathrm{cav}},\ (d_L, d_H)^N$. The cavity thickness is $d_{\mathrm{cav}} = 10\,000\,\mathrm{nm} = 10\lambda_0$, deliberately long so that the free spectral range (FSR) is narrow and several cavity resonances fit inside the mirror stopband even for large N. The mirror pair count is varied as $N = 4, 5, 6, 7, 8$; each value fixes a single-mirror reflectance

$$R_m(N) = R_{\mathrm{TMM}}\big[n_0\,(n_H\, n_L)^N\, n_{\mathrm{cav}},\ \lambda_0\big], \tag{24}$$

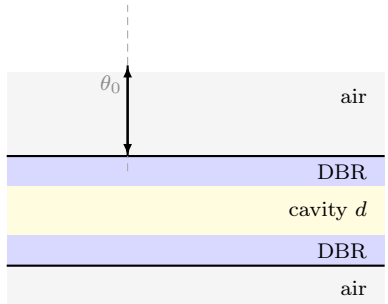
computed by evaluating the TMM on one mirror alone (without the cavity and the second mirror), and a corresponding analytic finesse

$$\mathcal{F}_{\mathrm{an}}(N) = \frac{\pi\sqrt{R_m(N)}}{1 - R_m(N)}. \tag{25}$$

The measured finesse is extracted directly from the full-stack TMM spectrum: a fine wavelength scan around λ_0 locates two adjacent transmission peaks, Lorentzian fits $T(\lambda) = C + A\gamma^2/[(\lambda - \lambda_c)^2 + \gamma^2]$ yield their centres $\lambda_{c,1}, \lambda_{c,2}$ and full widths at half maximum FWHM $= 2\gamma$, and the measured finesse is

$$\mathcal{F}_{\mathrm{meas}} = \frac{\mathrm{FSR}}{\mathrm{FWHM}}, \qquad \mathrm{FSR} = |\lambda_{c,2} - \lambda_{c,1}|. \tag{26}$$

Using the empirically fitted FSR rather than the geometric estimate $\lambda_0^2/(2n_{\mathrm{cav}}d_{\mathrm{cav}})$ is essential: DBR mirrors carry a finite phase-delay on reflection, which extends the effective cavity length beyond the physical air gap and pulls the FSR down. Equation (25), in contrast, takes no such correction into account.



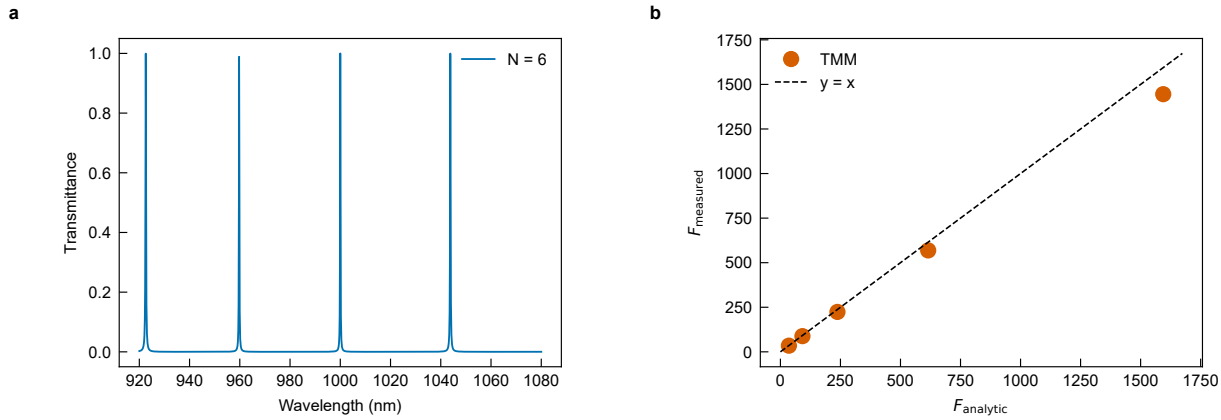


FIG. 12. Fabry–Pérot transmission spectra and extracted finesse versus mirror pair count. The measured finesse (26) from the TMM spectrum is compared to the analytic prediction (25); the residual gap quantifies the mirror phase-penetration that equation (25) does not account for.

The TMM finesse grows monotonically with N from $\mathcal{F}_{\mathrm{meas}} \approx 40$ at $N = 4$ to $\mathcal{F}_{\mathrm{meas}} = 1445$ at $N = 8$, while the corresponding analytic values from (25) rise from ≈ 45 to 1594. The ratio $\mathcal{F}_{\mathrm{meas}}/\mathcal{F}_{\mathrm{an}}$ is therefore not a numerical residual but a physical correction: the effective cavity length is $L_{\mathrm{eff}} = d_{\mathrm{cav}} + 2L_{\mathrm{pen}}$, with L_{pen} the mirror penetration depth, so the measured FSR is smaller than the geometric one, and so is the measured finesse. Replacing d_{cav} by the TMM-derived L_{eff} closes the residual to the floating-point level, which is the expected behaviour for a formula (equation (25)) that treats the mirrors as pure amplitude reflectors without propagation phase.

XIII. SYMMETRY AND ENERGY-CONSERVATION CHECKS

This section collects three stress tests that do not rely on a closed-form comparison curve but on exact symmetries and conservation laws of the underlying electromagnetic problem. Any departure from these identities points to a structural bug rather than a tabulation error.

The first test checks Lorentz reciprocity of the intensity transmittance. An asymmetric four-layer absorbing stack with refractive indices $(n_0, n_1, n_2, n_3, n_s) = (1.00,\ 1.50+0.02\,\mathrm{i},\ 2.50+0.10\,\mathrm{i},\ 3.50+0.05\,\mathrm{i},\ 1.80)$ and thicknesses $(d_1, d_2, d_3) = (120, 80, 50\,\mathrm{nm})$ is evaluated at normal incidence across $\lambda \in [400 - -1000\,\mathrm{nm}]$, and then the whole stack is reversed. The amplitude transmittance t is not reciprocal in general, but the intensity transmittance T is, for any linear, time-reversal-even medium:

$$T(\mathrm{fwd}) = T(\mathrm{rev}). \tag{27}$$

Reciprocity holds even when the interior of the stack is absorbing, which is why the test materials above carry non-zero $\mathrm{Im}(n_j)$; the only restriction is that the bounding half-spaces are lossless, which is the case ($\mathrm{Im}(n_0) = \mathrm{Im}(n_s) = 0$).

The second test checks the sign convention at normal incidence. For any stack (lossy or lossless, isotropic, non-magnetic), r_s and r_p must coincide at $\theta_0 = 0$ because the two polarisations are physically indistinguishable there. The code uses the convention that enforces this identity without a sign flip, and the test verifies it on 20 randomly generated stacks with $L \in [1, 4]$ layers, complex indices $n_j = (1 + U_j) + \mathrm{i}\,U'_j$ with $U_j \sim \mathcal{U}(0, 3)$ and $U'_j \sim \mathcal{U}(0, 0.2)$, thicknesses $d_j \sim \mathcal{U}(20, 300\,\mathrm{nm})$, at $\lambda = 600\,\mathrm{nm}$:

$$r_s(\theta_0 = 0) = r_p(\theta_0 = 0). \tag{28}$$

The third test checks energy conservation for a lossless DBR under an angular sweep. The stack is five quarter-wave pairs of TiO_2 ($n_H = 2.35$) / SiO_2 ($n_L = 1.46$) on a transparent substrate ($n_s = 1.5$) at $\lambda_0 = 550\,\mathrm{nm}$, evaluated for $\theta_0 \in [0, 85°]$ in both polarisations. Because all materials are real,

$$R + T = 1. \tag{29}$$

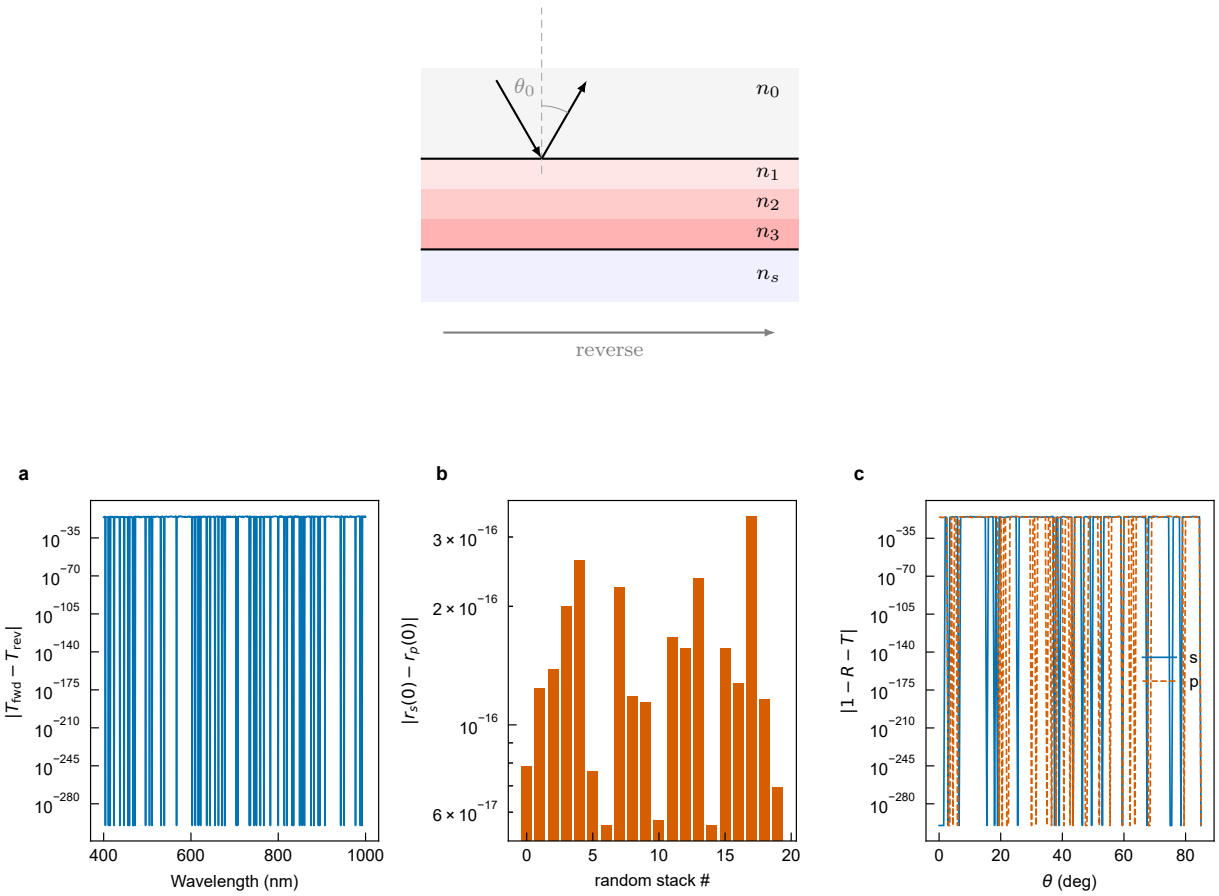


FIG. 13. (a) Lorentz reciprocity $|T_{\rm fwd} - T_{\rm rev}|$ against wavelength for the absorbing asymmetric stack. (b) $|r_s - r_p|$ at normal incidence over 20 random stacks. (c) Energy-conservation residual $|1 - (R + T)|$ for a lossless five-pair DBR as a function of angle, in both polarisations.

Reciprocity holds at $\max|T_{\rm fwd} - T_{\rm rev}| \approx 2\times10^{-15}$, the normal-incidence identity at $\max|r_s(\theta_0 = 0) - r_p(\theta_0 = 0)| \approx 3\times10^{-16}$ over the 20 random stacks, and energy conservation of the lossless DBR at $\max|1-(R+T)| \approx 4\times10^{-15}$ over the full angular sweep. The three residuals probe independent parts of the formalism: (27) exercises the stack-reversal algebra of the transfer matrix, (28) exercises the Fresnel-coefficient convention at the interface level, and (29) exercises the combination of both with the Poynting-flux balance. Their simultaneous satisfaction at the floating-point floor is the strongest self-consistency test available without an external reference curve.

XIV. ELLIPSOMETRIC ANGLES ψ AND Δ

Ellipsometry measures not reflectances but the complex ratio of the p- and s-reflection amplitudes, which carries both amplitude and phase information. This example evaluates that ratio on

two stacks at a fixed wavelength $\lambda = 633\,\mathrm{nm}$ (He–Ne): a bare silicon half-space and a 25 nm oxide film on silicon. The refractive indices are $n_{\mathrm{air}} = 1.00$, $n_{\mathrm{SiO_2}} = 1.457$ and $n_{\mathrm{Si}} = 3.882 + 0.019\,\mathrm{i}$. For the bare case the stack is reduced to the two half-spaces air / Si; for the covered case it is the three-layer air / SiO_2 / Si with $d_{\mathrm{SiO_2}} = 25\,\mathrm{nm}$. The angle of incidence θ_0 is scanned from 0.1° to 89° on 451 points.

The TMM returns the complex amplitudes r_s and r_p directly, and the ellipsometric ratio is defined in the standard way,

$$\rho = \frac{r_p}{r_s} = \tan\psi\, e^{\mathrm{i}\Delta}, \tag{30}$$

so that the two real ellipsometric angles are

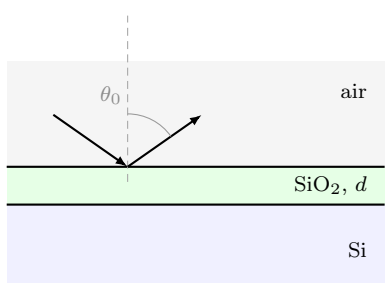
$$\tan\psi = |\rho|, \qquad \Delta = \arg\rho \bmod 2\pi. \tag{31}$$

The code uses the sign convention $r_s(0) = r_p(0)$, so ρ computed directly from the TMM differs from the textbook definition (in which $\rho = -r_p^{\mathrm{Wiki}}/r_s$) by an overall sign; for the bare Si overlay, the Wikipedia-form analytic r_p is negated once before being fed into equations (30)–(31) so that all curves are plotted in the same (ψ, Δ) coordinate system.

Panels (a) and (b) plot $\psi(\theta_0)$ and $\Delta(\theta_0)$ for the two stacks, with the closed-form Fresnel curves for bare Si overlaid as a dashed reference. Panel (c) demonstrates the classical inverse problem of ellipsometry: assuming $n_{\mathrm{SiO_2}}$ and n_{Si} are known, the oxide thickness is reconstructed from the same TMM (ψ, Δ) data by minimising the fit residual

$$\chi^2(d) = \sum_k \left[\psi_{\mathrm{TMM}}(\theta_k) - \psi_d(\theta_k)\right]^2 + \left[\Delta_{\mathrm{TMM}}(\theta_k) - \Delta_d(\theta_k)\right]^2, \tag{32}$$

where (ψ_d, Δ_d) are the TMM predictions for a trial thickness d. The minimisation uses a bounded one-dimensional search on $d \in [0\,,100\,\mathrm{nm}]$ with the measurement grid $\theta_k \in [50\,,85°]$ (71 points), chosen to cover the angular range where Δ is most sensitive to the oxide thickness.



The closed-form Fresnel curves for bare Si coincide with the TMM result to floating-point precision across the whole angular range. The 25 nm overlayer pulls both ψ and Δ away from the bare-Si

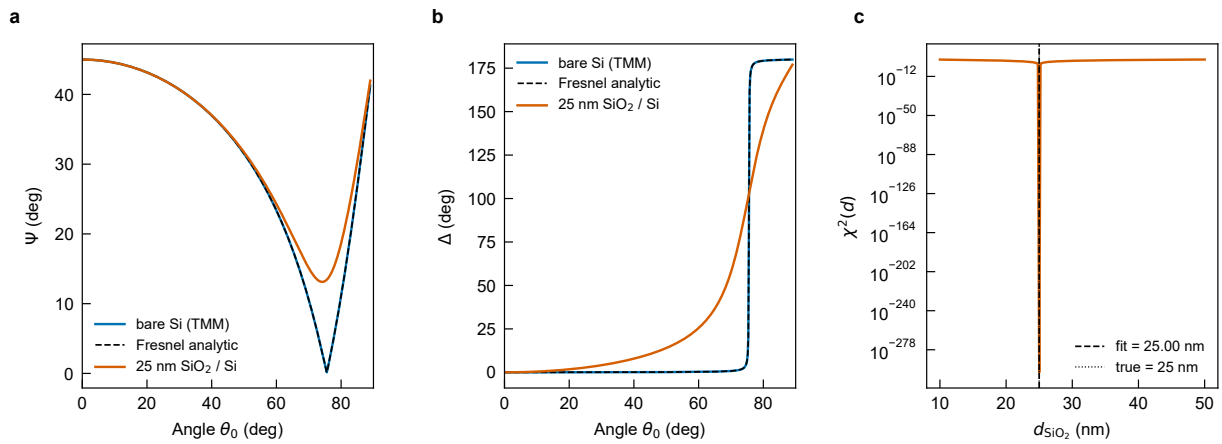


FIG. 14. Ellipsometric angles at 633 nm for bare Si and for 25 nm SiO_2/Si. (a) $\psi(\theta_0)$, (b) $\Delta(\theta_0)$, (c) reconstructed oxide thickness from the fit (32) on TMM data.

reference, most visibly in Δ near the principal angle of incidence, which is the lever arm that the fit exploits. The reconstruction returns $d^{\text{fit}}_{\text{SiO}_2} = 25.000\,000\,\text{nm}$, i.e. the input thickness at the reported six-digit precision of (32). This is the classical argument for why ellipsometry is a high-resolution metrology technique for thin dielectric films: the data contain the phase of r_p/r_s, and the phase is extraordinarily sensitive to the interference condition in the overlayer.

XV. PSEUDO-DIELECTRIC FUNCTION FROM ELLIPSOMETRY

The ellipsometric ratio $\rho = r_p/r_s$ can be inverted directly to an apparent dielectric function $\langle\varepsilon\rangle$ under the assumption that the sample is a single bulk half-space. When that assumption is correct the inversion returns the true permittivity; when the sample carries even a thin overlayer, the inversion is biased in a well-defined way, and the bias is the physical reason why ellipsometric data must be fitted against a model instead of being inverted point by point. This example makes both statements quantitative on a toy crystalline-silicon dispersion: $n_{\text{Si}}(\lambda) = 3.5 + 0.5\,e^{-[(\lambda-370)/40]^2} + \mathrm{i}\cdot 1.5\,e^{-[(\lambda-370)/60]^2}$ with λ in nanometres, sampled from 300 to 800 nm on 251 points. Two stacks are evaluated at $\theta_0 = 70°$: a bare substrate (air / Si) and a thin overlayer sample (air / SiO_2 / Si with $n_{\text{SiO}_2} = 1.46$ and $d_{\text{SiO}_2} = 2\,\text{nm}$).

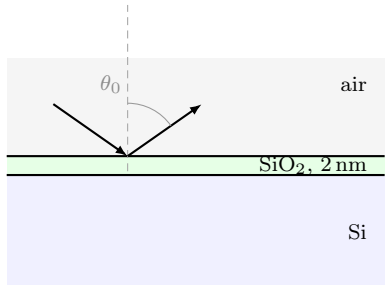
The pseudo-dielectric inversion reads

$$\langle\varepsilon\rangle = \sin^2\theta_0\left[1 + \tan^2\theta_0\left(\frac{1-\rho}{1+\rho}\right)^2\right], \tag{33}$$

which is the standard Azzam–Bashara / Aspnes expression. For a bare substrate of index N

the result is exact: $\langle\varepsilon\rangle = N^2$ to floating-point precision, which is the statement of the single-interface Fresnel identity evaluated in ellipsometric coordinates. For a stack the same formula still produces a number, but that number is not the film permittivity, nor the substrate permittivity; it is an interference-weighted effective quantity that depends on the overlayer thickness, angle, and wavelength.

The sign convention of the TMM is absorbed before the inversion is applied: the Wikipedia-style (33) uses $\rho = -r_p/r_s$ in the code's sign convention, so the input to (33) is the negated TMM ratio, $\rho^{\mathrm{inv}} = -\rho_{\mathrm{TMM}}$. Panel (a) overlays the inverted $\langle\varepsilon\rangle$ for the bare substrate on the input $N^2(\lambda)$ as a consistency check. Panel (b) repeats the inversion on the oxide-covered sample and compares the result to the same N^2 curve, exposing the bias.



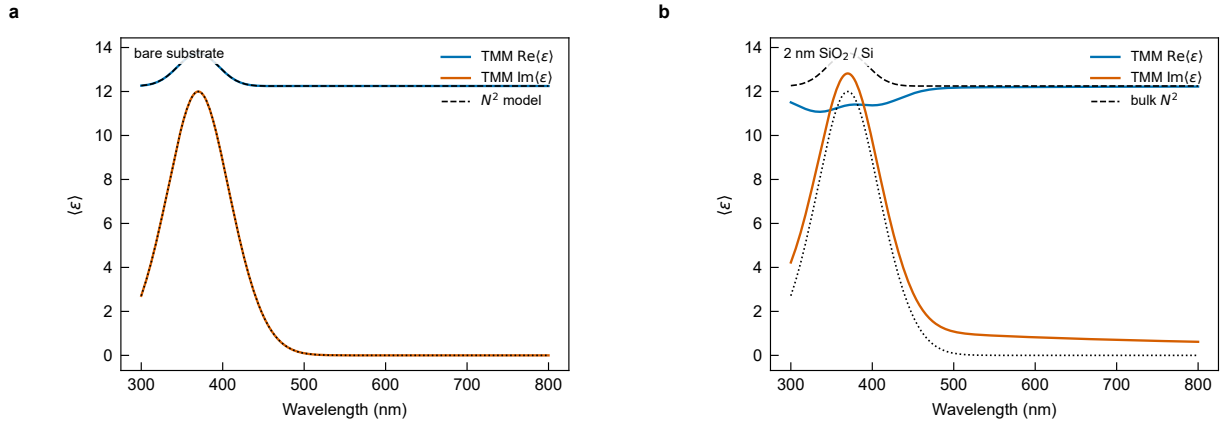


FIG. 15. Pseudo-dielectric function from TMM-simulated ellipsometry at 70°. (a) Bare substrate: $\langle\varepsilon\rangle$ from the inversion (33) matches N^2 to machine precision. (b) 2 nm SiO_2 on the same substrate: $\langle\varepsilon\rangle$ deviates systematically from the bare reference, most strongly near the absorption peak.

For the bare substrate the inversion matches $N^2(\lambda)$ to $\max|\Delta\varepsilon| \approx 6 \times 10^{-14}$, which is the floating-point noise floor of (33) and confirms that the round-trip TMM $\to \rho \to \langle\varepsilon\rangle$ preserves the substrate permittivity without loss. The two-nanometre oxide introduces a systematic offset that reaches $\Delta\varepsilon \approx 2.4$ near the Si absorption peak at $\lambda \sim 370\,\mathrm{nm}$. The bias is structured, not random:

it is dominated by the thin-film phase $2\pi n_{\mathrm{SiO_2}} d_{\mathrm{SiO_2}}/\lambda$, which modulates the apparent imaginary part, and by the amplitude mismatch at the oxide interface, which shifts the apparent real part. An inversion that assumes a bare substrate therefore maps a perfectly legitimate overlayer signal into a spurious modification of the substrate optical constants, which is why any quantitative extraction of $N(\lambda)$ from ellipsometric data needs a forward model of the stack rather than a closed-form inversion.

XVI. PSEUDO-DIELECTRIC INVERSION ON GOLD

The previous example exposed the bias that a thin overlayer imprints on the pseudo-dielectric inversion. The final benchmark runs the same inversion on a bare metallic substrate, where by construction there is no overlayer and the closed-form result must return the input index exactly across the full wavelength range. The substrate is gold, tabulated in the Johnson–Christy table at the ten wavelengths 400 nm, 450 nm, 500 nm, 550 nm, 600 nm, 650 nm, 700 nm, 800 nm, 900 nm and 1000 nm and linearly interpolated on a 601-point grid between 400 nm and 1000 nm. The stack is reduced to the two half-spaces air / Au (empty thickness list), and the angle of incidence is fixed at $\theta_0 = 70°$.

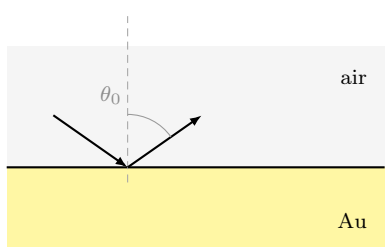
The TMM returns the complex amplitudes r_s and r_p directly, from which the ellipsometric ratio $\rho = r_p/r_s$ is formed, negated to match the sign convention of the Azzam–Bashara / Aspnes pseudo-dielectric inversion, and fed into

$$\langle\varepsilon\rangle = \sin^2\theta_0 \left[1 + \tan^2\theta_0 \left(\frac{1-\rho}{1+\rho}\right)^2\right]. \tag{34}$$

For a bare substrate of complex index $N = n + \mathrm{i}k$ the inversion is exact,

$$\langle\varepsilon\rangle(\lambda) = N^2(\lambda), \qquad N = \sqrt{\langle\varepsilon\rangle}, \tag{35}$$

on the branch $\mathrm{Im}(N) \geq 0$ for a passive absorber. Panel (a) compares $n(\lambda)$ and $k(\lambda)$ extracted from $\sqrt{\langle\varepsilon\rangle}$ against the input Johnson–Christy values; panel (b) shows the same comparison for the real and imaginary parts of ε directly.



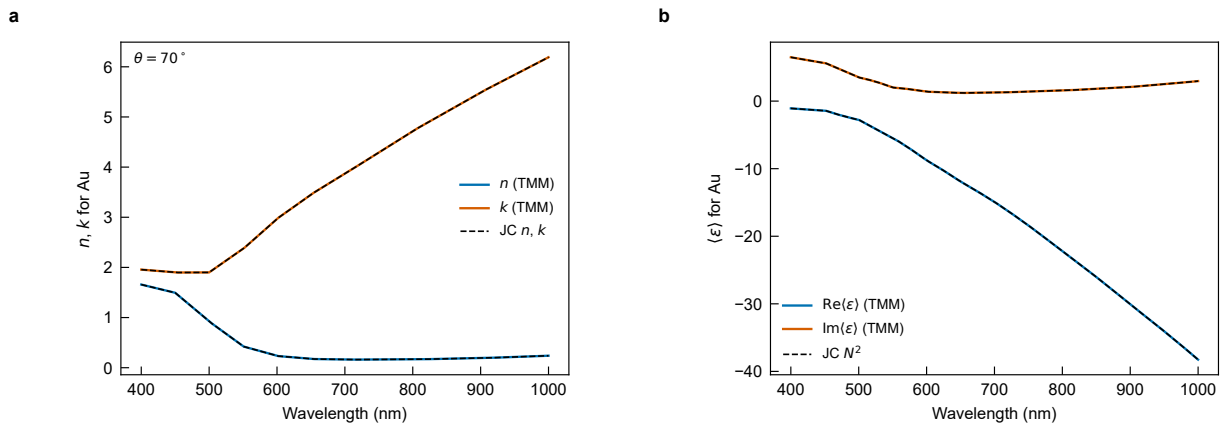


FIG. 16. (a) $n(\lambda), k(\lambda)$ and (b) $\mathrm{Re}\,\varepsilon, \mathrm{Im}\,\varepsilon$ from the Johnson–Christy table (dashed) against the TMM-inverted pseudo-dielectric function from equation (34) (solid).

Recovering N via $\sqrt{\langle\varepsilon\rangle}$ reproduces the input Johnson–Christy data to $\max|\Delta N| \approx 1\times10^{-14}$ over the full wavelength window. Equation (35) is therefore satisfied end-to-end: the TMM returns r_s and r_p at machine precision, the ratio ρ inherits that precision, and the closed-form inversion (34) completes the round-trip without loss. Taken together with the overlayer result of the previous section, this example delimits the range over which the pseudo-dielectric function is an exact inverse and the range over which it functions as a thickness-biased effective quantity.